Food Waste Prediction: A Supervised Regression Study

Fahmidul Islam, 23201662
Md. Mushfiqur Rahman, 23201471

***Abstract*—Food waste is a critical sustainability and economic challenge for the food service industry. This paper presents a supervised regression study that builds and compares three machine learning models—Linear Regression, Decision Tree Regressor, and a Multi-Layer Perceptron (MLP) Neural Network—to predict daily food waste (in kilograms) from operational and environmental features. The dataset comprises 911 daily kitchen records spanning January 2022 to September 2024, covering variables such as meals served, kitchen staff count, temperature, humidity, food category, and staff experience level. An unsupervised KMeans clustering analysis is additionally performed to uncover latent structure in the data. Model performance is evaluated using R^2, MSE, RMSE, and MAE metrics with 10-fold cross-validation. The Neural Network achieves the best generalisation (R^2 = 0.83, RMSE = 9.32 kg), followed by the Decision Tree (R^2 = 0.82, RMSE = 9.83 kg) and Linear Regression (R^2 = 0.75, RMSE = 11.37 kg). Results confirm that *meals_served* and *past_waste_kg* are the dominant predictors of food waste, and that the feature space contains genuine latent structure recoverable without labels.**

***Index Terms*—food waste prediction, supervised regression, decision tree, neural network, MLP, KMeans clustering, feature engineering, sustainability**

I. Introduction

Food waste is a global sustainability and economic problem of significant scale. Approximately one-third of all food produced globally is wasted, generating avoidable greenhouse gas emissions and imposing substantial financial costs on the food service industry [1].

Accurate prediction of daily food waste enables kitchen managers to optimise inventory, adjust preparation quantities, and reduce both operational costs and environmental impact. Machine learning regression models are well-suited to this task, as they can identify complex, non-linear relationships among operational and environmental variables that influence waste generation.

This paper investigates three supervised regression algorithms applied to a kitchen operational dataset: Linear Regression as an interpretable baseline, a Decision Tree Regressor, and a Multi-Layer Perceptron (MLP) Neural Network. An unsupervised KMeans clustering analysis is additionally performed to explore natural groupings in the data without using target labels.

The contributions of this study are: (1) a thorough exploratory data analysis and pre-processing pipeline for a real kitchen waste dataset; (2) a systematic comparison of three regression models using four evaluation metrics and 10-fold cross-validation; and (3) unsupervised insights that confirm genuine latent structure in the feature space.

II. Dataset Description

A. Overview

The dataset contains daily operational records from a kitchen environment, covering January 2022 to September 2024. A labelled training set (911 samples) and an external test set (911 samples) are provided, mirroring a competition-style evaluation setup. The target variable *food_waste_kg* is continuous; hence this is treated as a regression problem. Table I summarises the dataset properties.

TABLE I
Dataset Properties

Property	**Detail**
Total features	11 (10 input + 1 target)
Training samples	911 rows
Test samples	911 rows
Problem type	Regression (continuous)
Date range	Jan 2022 – Sep 2024

B. Feature Types

The ten input features span quantitative, binary, categorical, and date-derived types, as summarised in Table II. Categorical features *staff_experience* and *waste_category* require label encoding; the *date* column is decomposed into month, year, and quarter.

TABLE II
Feature Summary

Feature	**Type**	**Description**
meals_served	Quantitative	Meals prepared/served
kitchen_staff	Quantitative	Staff on duty
temperature_C	Quantitative	Ambient temp. (°C)
humidity_percent	Quantitative	Relative humidity (%)
day_of_week	Quantitative	0 (Mon) – 6 (Sun)
special_event	Binary	1 = event day
past_waste_kg	Quantitative	Prior-day waste (kg)
staff_experience	Categorical	Beginner/Interm./Expert
waste_category	Categorical	Dairy/Grains/Meat/Veg.
date	Date	→ month, year, quarter

C. Correlation Analysis

Pearson, Spearman, and Kendall correlations were computed for all numerical features. Key findings are as follows:

- *meals_served* has the highest positive correlation with the target, confirming that preparing more meals directly increases waste.

- *past_waste_kg* shows a meaningful correlation, indicating temporal persistence in waste patterns across consecutive days.
- *temperature_C* and *humidity_percent* exhibit weak but consistent positive correlations, suggesting warmer and more humid conditions may accelerate spoilage.
- *kitchen_staff* and *day_of_week* show near-zero correlation with the target, confirming they are poor individual predictors.
- No two input features exceed 0.80 mutual correlation, indicating no severe multicollinearity.

D. Target Distribution

The target *food_waste_kg* is right-skewed. As shown in Table III, 65.3% of records fall in the 25–50 kg range, while only 2.5% exceed 100 kg, corresponding to rare special-event days.

TABLE III
TARGET VARIABLE DISTRIBUTION (BINNED)

Range	Share (%)	Observation
0 – 25 kg	9.55	Low-waste days
25 – 50 kg	65.31	Most common
50 – 75 kg	22.50	Moderate waste
75 – 100 kg	0.10	High-waste days
>100 kg	2.52	Outlier events

III. DATA PRE-PROCESSING

A. Dirty Categorical Values

Both categorical columns contained case-inconsistent entries representing the same value: *waste_category* included entries such as 'MeAt' and 'MEAT' (5 apparent values collapsing to 4 true categories), while *staff_experience* had 'intermediate', 'Intermediate', and 'INTERMEDIATE' (4 apparent values collapsing to 3 true levels). Both columns were normalised to uppercase using `.str.strip().str.upper()` prior to label encoding.

B. Missing Values

The *staff_experience* column contained 164 missing values ($\approx$18% of training records). Row deletion would have significantly reduced training set size and introduced sampling bias. Mode imputation per *waste_category* group was applied to preserve the full training set.

C. Feature Scaling

Input features span vastly different numeric ranges: *meals_served* reaches approximately 4,730 while *special_event* is binary (0/1). `StandardScaler` (zero mean, unit variance) was applied to all features before training Linear Regression and the Neural Network. The Decision Tree is scale-invariant and was trained on unscaled data. Critically, the scaler was fitted only on training data and applied to validation and test sets, preventing data leakage.

D. Additional Pre-Processing Steps

The *date* column was decomposed into month, year, and quarter features then dropped. The non-predictive ID column was also dropped. `LabelEncoder` was applied to *staff_experience* and *waste_category*, with encoders fitted on training data and reused on the test set to ensure consistent encoding.

IV. DATASET SPLITTING

An 80/20 random split of the training file was used for model development (Table IV). The 20% slice served as an internal validation set for model comparison and hyperparameter tuning. The external test set was kept completely sealed throughout all training and tuning decisions, equivalent to a competition holdout. A stratified split was not required since the target is continuous. All splits used `random_state=42` for reproducibility.

TABLE IV
DATASET SPLITS

Split	Size	Purpose
Train (80%)	729 samples	Model fitting
Validation (20%)	182 samples	Tuning & comparison
External test	911 samples	Final evaluation

V. MODEL TRAINING & TESTING

A. Linear Regression

Ordinary Least Squares (OLS) Linear Regression was trained on `StandardScaler`- normalised features as a simple and interpretable baseline. It fits a linear combination of input features to minimise the sum of squared residuals. Top-magnitude coefficients correspond to *meals_served* and *past_waste_kg*, consistent with the correlation analysis.

B. Decision Tree Regressor

A CART Decision Tree was trained on unscaled features with `max_depth=8` and `min_samples_leaf=5` to prevent overfitting. The tree recursively partitions the feature space and assigns the mean target value of training samples in each leaf as the prediction. Built-in feature importance scores provide interpretability.

C. Neural Network (MLP Regressor)

A Multi-Layer Perceptron with architecture Input(13)$\rightarrow$128$\rightarrow$ 64$\rightarrow$32$\rightarrow$Output(1), ReLU activations, and Adam optimiser (learning rate 0.001) was trained with early stopping (validation fraction 0.1, max 500 iterations). The training loss curve was monitored to confirm convergence.

D. KMeans Clustering (Unsupervised)

KMeans clustering ($k=4$, selected via the Elbow Method over $k=2$–10, `n_init=10`) was applied to the scaled training features to explore natural groupings without target labels. Cluster structure was visualised via scatter plots, a PCA 2D projection, and boxplots of *food_waste_kg* per cluster.

E. Cross-Validation

10-fold cross-validation was applied to all three supervised models on the training data to obtain robust performance estimates that are less sensitive to any specific train/validation split.

VI. Results and Model Comparison

A. Validation Set Performance

Table V presents the validation set metrics for all three supervised models. The Neural Network achieves the best performance across all four metrics, confirming that non-linear feature interactions are significant in this dataset.

TABLE V
Model Comparison — Validation Set

Model	R^2	MSE	RMSE (kg)	MAE (kg)
Linear Regression	0.748	129.34	11.37	5.50
Decision Tree	0.818	96.67	9.83	6.08
Neural Network	0.830	86.91	9.32	5.78

B. Feature Importance

Decision Tree importance scores confirm that *meals_served* and *past_waste_kg* are the two most informative features, aligning with the Pearson correlation analysis. Date-derived features (month, quarter) and *waste_category* contribute moderately; *kitchen_staff* and *day_of_week* contribute least.

C. Residual Analysis

Residual distributions for all three models are approximately centred near zero, indicating no systematic bias in the central portion of the target distribution. All models tend to underestimate on extreme high-waste days (>100 kg) due to the sparsity of such outlier examples in training data.

D. KMeans Clustering Insights

With $k=4$ clusters, boxplots of *food_waste_kg* per cluster show clearly separated median waste levels, confirming genuine latent structure in the feature space. One cluster consistently captures high-*meals_served* days with elevated prior waste; special-event days tend to group separately. The PCA 2D projection shows partially overlapping but broadly separable clusters, reflecting the inherent continuity of the regression target.

VII. Discussion

All three supervised models achieve $R^2 > 0.74$, explaining more than 74% of variance in daily food waste from the available features. The remaining unexplained variance likely reflects human behaviour patterns and external factors absent from the dataset, such as menu complexity, ingredient freshness, and cultural events.

The Neural Network's advantage over the Decision Tree is modest ($\Delta R^2 \approx 0.01$), suggesting that while non-linear interactions are present, a depth-constrained tree captures most of the predictable structure at lower computational cost and with greater interpretability—a compelling trade-off for operational deployment where model transparency matters.

Linear Regression's underperformance confirms that the relationship between operational features and food waste is non-linear in nature, a finding reinforced by the significant feature importances attributed to interaction-sensitive features in the tree model.

VIII. Conclusion

This paper presented a comparative supervised regression study for predicting daily food waste in a kitchen environment. Three machine learning models were trained, tuned, and evaluated on 911 daily operational records spanning 2022–2024. The Neural Network ($R^2 = 0.83$, RMSE $= 9.32$ kg) outperformed the Decision Tree ($R^2 = 0.82$, RMSE $= 9.83$ kg) and Linear Regression ($R^2 = 0.75$, RMSE $= 11.37$ kg). *meals_served* and *past_waste_kg* emerged as the dominant predictors across all analyses. KMeans clustering ($k=4$) confirmed that operational factors—meals served, past waste, and special events—are the primary axes of variation in daily food waste behaviour, and that this structure is recoverable without labels.

Future work should investigate ensemble methods (Gradient Boosting, Random Forest) and temporal models (LSTM) to better capture sequential dependencies and improve prediction accuracy on high-waste outlier days.

Code Availability

The source code for this project is publicly available on GitHub:
https://github.com/s-h-u-v-o/CSE422_Artificial-Intelligence/tree/main/Lab%20Project

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